

Final Project

Proposal due Monday 07/27 at 9:00am (EST)

Project due Tuesday 08/04 at 9:00am (EST)

Introduction

This final project is meant to act as a summary of the skills you’ve learned throughout this course, including both fundamental programming skills, asking computationally-solvable questions, using AI chatbots as programming tools, and writing about code. However, it’s also meant to be fun and interesting! I hope you take this as a chance to ask a question that interests you about a subject you’re excited about, and I’m excited to see what you come up with!

There will be three parts to this project. In place of an Assignment 5, you will instead be asked to write a project proposal due the morning of Monday, July 27 at 9:00am (EST). Then, on the morning of the course final exams — Tuesday, August 4 at 9:00am (EST) — you will submit your analysis code and a write up of what your analysis revealed about your research question.

Project Requirements

For your final project, you will design a research project that uses Python code to analyze a real-world dataset. You may either choose one of the datasets linked below, or find and use your own dataset.

- [Stack Overflow Annual Developer Survey](#)
- [SAT Scores by State \(CORGIS\)](#)
- [College Majors & Earnings \(FiveThirtyEight\)](#)
- [New Orleans Slave Sales, 1856–1861 \(CORGIS\)](#)
- [Broadway Show Data \(CORGIS\)](#)
- [Tate Museum Artwork Collection](#)
- [FBI Crime Data by US State \(CORGIS\)](#)
- [NOAA Weather Data \(CORGIS\)](#)
- [USGS Earthquake Feed](#)
- [Coffee Quality Ratings \(CORGIS\)](#)
- [Global Smoking Trends \(CORGIS\)](#)

- [Astronauts Dataset \(CORGIS\)](#)

Note that many of the datasets above were found by Claude — you can try using the AI chatbot of your choice to look for datasets on a particular topic if you're interested! A lot of these datasets also come from [CORGIS](#), which is a great place to look for more options!

The code you write must meet the following specifications:

- Use for and / or while loops
- Use conditional statements
- Include ≥ 3 functions
- Use at least 2 of the following data structures: lists, dictionaries, tuples, sets
- Read in your data from an external file
- Make significant use of NumPy arrays and / or Pandas dataframes
- Use built-in NumPy and / or Pandas functionality to streamline some part of your data analysis
- Use Matplotlib to create ≥ 3 figures that contribute to answering your research question

You will be allowed to use AI tools to help write your analysis code. However, the majority should be written by you independently; I will ask you to submit all chatbot logs and mark in your code what was generated.

Proposal

due 07/27 @ 9:00am (EST)

Your final project proposal should be a 300–400 word write-up that includes a description of your research question, how you will solve this question computationally, and how your analysis code will fulfill the requirements detailed above.

Specifically, you should describe in detail the research question you will ask, why it is interesting, what you hope to learn from it, and why your data will allow you to answer this question. You should include what dataset you will use in your analysis; if it isn't one of the datasets listed in this document, you should submit a copy of it with your proposal. Then, you should describe how you will answer this question computationally. This description should include how you will fulfill each of the specified code requirements and which parts you will write by hand and which you plan to use an AI chatbot to write.

Expected Submission(s): Your written proposal (can be .docx, .pdf, etc) and dataset (if you've found your own).

Final Project

due 08/04 @ 9:00am (EST)

Your final project submission will consist of several parts.

First, you should submit your analysis code. This code should meet all of the listed requirements. It should also include informative comments and indicators of which code was AI generated.

Additionally, you should submit a 200–400 word discussion of your research question and what you learned from your analysis. You should embed and discuss the figures generated by your code.

Finally, you should also submit a `final-project-logs.txt` file that includes all chatbot conversations you had about this project.

Expected Submission(s): Your code (.py), your written analysis (any text form), and chatbot logs (.txt file).